

Able Thomas

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SUMMARY

Second-year Computer Science student (CGPA: 9.75/10) at VIT Vellore with a strong foundation in data structures, algorithms, databases, and backend development. Builds projects spanning statistical modeling, machine learning, and system design, applying algorithmic problem-solving to practical engineering challenges.

EDUCATION

Vellore Institute of Technology (VIT), Vellore — B.Tech, Computer Science and Engineering

Expected 2029 | CGPA: 9.75/10

- 1st Rank Holder, CSE Cybersecurity, Academic Year 2025-26
- Merit Scholarship Recipient
- ICSE Board Examination — 98.4%;
- ISC Board Examination-96.75%

TECHNICAL SKILLS

Languages: C++, C, Python, SQL

CS Fundamentals: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Computer Networks

Backend & Databases: MySQL, REST APIs

Data / ML: Statistical Analysis, Bayesian Inference, Machine Learning Fundamentals, Streamlit

Tools: Git, GitHub, VS Code, Flutter, React, HTML, CSS

PROJECTS

1) MiniPlay – Full-Stack Music Streaming Web Application | Completed

Tools- HTML, CSS, JavaScript, PHP, MySQL, XAMPP

- Developed a full-stack music streaming platform with user authentication, music discovery, genre and album browsing, search, liked songs, and personalized playlists.
- Designed and implemented a relational MySQL database to manage users, artists, albums, genres, songs, playlists, and playlist-song relationships.
- Implemented backend functionality using PHP for secure database interaction, user management, music retrieval, playlist operations, and dynamic content rendering.
- Built a responsive web interface for browsing and playing music with structured navigation across artists, albums, genres, playlists, and user preferences.

2) ResQ – Smart Ambulance & SafeRoute Routing System | In Progress

Tools- Flutter, Python, Flask, SUMO, Graph Algorithms, OpenStreetMap

- Developing a real-time intelligent routing platform integrating **Normal, SafeRoute, and Emergency Services** modes for context-aware urban navigation.
- Implementing graph-based shortest-path routing and SUMO traffic simulation to dynamically calculate ambulance routes from dispatch locations to patients and nearby hospitals.
- Designing a safety-aware routing engine that considers road characteristics, street lighting, nearby businesses, user reports, and time-of-day conditions for safer route selection.
- Built a Flutter-based mobile application and web-based emergency dashboard with ambulance dispatch, SOS alerts, route visualization, traffic simulation, and real-time ambulance tracking.